

ColorRef: Calibrating Natural-Language Feedback for Color Control

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Abstract

How can language help a model revise a color that is almost, but not quite, what is intended? We introduce COLORREF, a reference game in which a guesser predicts a color from a description, a teacher knows the hidden target and gives verbal feedback, and the guesser revises over successive rounds. The resulting trajectories let us study feedback as a control interface: a useful correction must communicate both which way to move and how far. Oracle feedback improves recorded-target agreement on 1,000 new descriptions, but controlled probes show that following the right direction can still produce excessive steps. We therefore calibrate graded phrases on separate colors and select wording according to the receiving model’s measured response. On the same 128 held-out starting colors, independently calibrated policies reduce one-step mean target error from 12.36 to 6.86 ΔE_{76} for Qwen3-14B and from 11.06 to 5.89 for Gemma 4 12B, relative to a fixed graded-wording rule. Supporting controls distinguish output representation, direction preservation, and downstream usefulness. ColorRef connects interactive reference to measurable correction behavior; its results concern recorded targets and prompt-conditioned control, rather than uniquely correct color meanings or human perceptual calibration.

1 Introduction

Color is a continuous perceptual domain that people describe through language. Some descriptions refer directly to color properties, such as *dark green* or *pale blue*; others draw on familiar objects, such as *mustard yellow*, or evoke associations, such as *abandoned hope*. These expressions offer different ways of communicating color, but their interpretation is not uniquely determined. A description can admit several plausible colors, and its intended referent can depend on context, experience, and subjective associations.

This relationship makes color a useful setting for studying how large language models interpret and use grounded language. Given a color description, a model can propose a color in a *one-shot prediction*, allowing us to examine its initial interpretation. Yet a single prediction captures only part of the communicative process. When a proposed color differs from what a speaker intends, the speaker can offer a correction—*a little darker* or *more blue*—and the listener can revise. This raises a complementary question: how effectively can a model use natural-language feedback to move from its initial interpretation toward an intended color?

We study this process through COLORREF, an interactive reference game with two roles (Figure 1). Each color description is paired with a recorded *target color*. The *guesser* receives only the description and proposes a *color guess*. The *teacher* knows the target, compares it with the guess, and provides natural-language *feedback*. The guesser then receives the description, its previous guess, and the latest feedback, and produces a revised guess. This exchange repeats until a stopping rule or revision budget ends the game; the target remains hidden from the guesser. The resulting trajectory makes it possible to examine initial interpretation, response to correction, and feedback production separately. We can therefore ask both whether a model follows a teacher’s feedback and whether model-produced feedback helps a receiver approach the recorded target.

Within this game, a verbal correction must be translated into a change in the current color guess. Two aspects of that change matter. *Direction* specifies which color property should increase or decrease, as in *lighter* or *more blue*. *Magnitude* specifies how large the change should be: *a little darker* and *much darker* request the same direction but different degrees of adjustment. Even when feedback leaves magnitude implicit, the guesser must choose a step size. Following the requested direction is

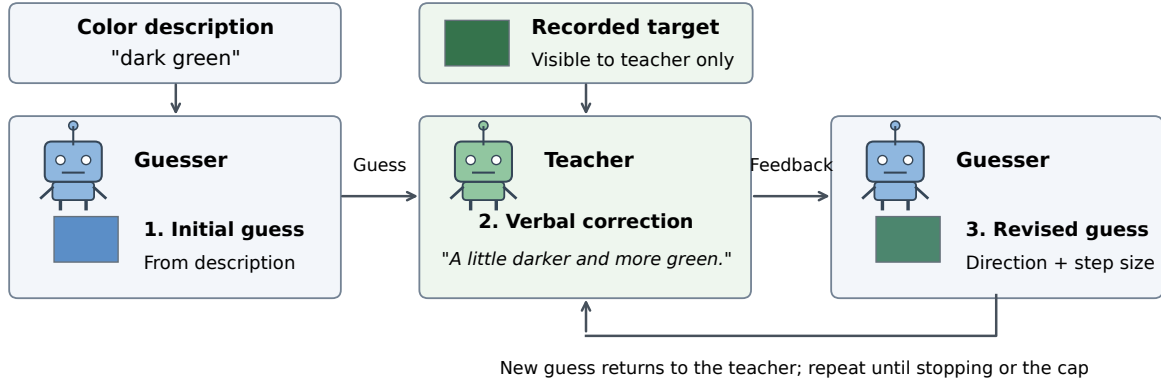


Figure 1: The ColorRef reference game, illustrated with *dark green*. Blue robots denote the guesser; the green robot denotes the teacher. The guesser sees the description and predicts a color; the teacher knows the hidden target and gives a correction. The revised guess becomes the state for the next exchange. A message such as *a little darker and more green* raises two questions: does the receiver move in the requested directions, and does it take an appropriate step? Swatches and wording here are schematic examples, not measured model responses.

therefore only part of a successful revision. A step can be too small to resolve the disagreement or so large that it overshoots the target. Graded modifiers make this problem especially informative: understanding their ordering need not mean that a receiver’s changes match the correction needed.

These distinctions motivate our research questions. Can a model revise its color guess in the directions requested by feedback? How does it interpret graded magnitude expressions, and can measuring those responses help a teacher choose more effective wording? On the production side, can a model faithfully express supplied corrections, and does preserving those corrections lead to useful revisions by the receiver? The first two questions concern *feedback following*; the third concerns *feedback generation* and its downstream effect. ColorRef lets us examine these behaviors separately while evaluating their consequences against the same recorded targets.

Language-to-color research studies prediction and reference among alternatives (McMahan and Stone, 2015; Monroe et al., 2017), while representation studies compare model embeddings with perceptual geometry (Abdou et al., 2021; Loyola et al., 2023). Comparative color terms have also been modeled as directions in color space (Winn and Muresan, 2018). Our behavioral focus complements these approaches by examining how corrections change the receiver’s current output. We measure agreement using ΔE_{76} , the Euclidean distance in CIELAB (International Commission on Illumination, 2018), without treating it as a strict

perceptual threshold or a judgment that alternative color meanings are invalid.

To answer these questions, we combine description-based reference games with controlled color-adjustment tasks. The games establish how initial interpretations change under teacher feedback; supplied-start comparisons and isolated adjustments help distinguish direction interpretation and step size from the difficulty of predicting a color from its description. We also compare model-produced feedback with deterministic oracle messages, examining both preservation of supplied corrections and their effect on a receiving guesser. Across these studies, agreement with the recorded target provides a common outcome, while movement and message diagnostics reveal why a correction may succeed or fail.

Our central contribution is *receiver-calibrated feedback*. We measure the changes induced by graded expressions such as *a little* and *much* for each correction direction on separate calibration colors. A teacher then selects the phrase whose typical displayed step best matches the remaining correction. This policy uses the same graded vocabulary as a fixed, unfitted wording rule, but bases its choice on the receiving model’s measured behavior. We evaluate independently calibrated policies for Qwen3-14B and Gemma 4 12B on the same 128 held-out starting colors.

The results connect initial interpretation, feedback following, and feedback generation. Oracle correction improves target agreement on 1,000 new color descriptions, showing that an initial guess

does not capture the full behavior available through interaction. Controlled adjustments show why correct direction and ordered magnitude responses can still be insufficient: large corrections can overshoot. Receiver calibration reduces held-out error for both models, with gains concentrated on avoiding excessive large steps. Teacher controls further show that faithfully restating supplied directions does not guarantee the best receiver accuracy. Together, these findings make ColorRef a setting for studying how language guides revision and how feedback can be adapted to its receiver. The evaluations concern model behavior against assigned targets; human interaction and creative image editing remain directions for future work.

2 Related Work

Color semantics and representation geometry. Color language has been studied through cross-linguistic regularities, perceptual geometry, and communicative efficiency (Berlin and Kay, 1991; Regier et al., 2007; Zaslavsky et al., 2018). Probabilistic and neural approaches map descriptions to perceptual observations (McMahan and Stone, 2015; Monroe et al., 2017). Abdou et al. (2021) compare color-term representations with CIELAB geometry; Loyola et al. (2023) study how alignment changes with subjectivity and abstraction. We complement representation probing with controlled output trajectories rather than infer internal representations from behavioral success.

Comparatives and interactive grounding. Winn and Muresan (2018) model comparative color terms as directions in color space. Reference games and pragmatic models treat communication as speaker–listener coordination (Lewis, 2008; Frank and Goodman, 2012; Goodman and Frank, 2016). Grounded clarification distinguishes perceptual and collaborative grounding (Benotti and Blackburn, 2021), while interactive evaluation studies common-ground formation (Imai et al., 2025). Our context is the current continuous state: we examine both directional corrections and the operational magnitude of graded modifiers.

Feedback and revision. Feedback supports instruction tuning, AI critique, and iterative refinement (Ouyang et al., 2022; Bai et al., 2022; Saunders et al., 2022; Madaan et al., 2023; Shinn et al., 2023). However, feedback and its reception can introduce separate errors. We use oracle-assisted

restatement and matched receiving revisions to distinguish supplied-direction preservation from usefulness, without equating linguistic audit scores with task success.

3 Framework

3.1 Descriptions and assigned targets

We construct the corpus from ColorNames-derived English descriptions and sRGB hex codes. After preprocessing and validity filtering, 518,830 rows remain. Each example (x_i, h_i, c_i, r_i) contains a description, recorded HEX target, its CIELAB coordinate, and a semantic-regime label. We retain description-validity scores ≥ 0.75 , including 43,187 rows at the boundary. The validity filter asks whether a string plausibly functions as a color description; it does not measure agreement about its paired target. Construction and filtering details appear in Appendix A.

The four regimes are *explicit grounded*, with direct color terms; *prototype mediated*, referring through objects or materials; *compound associative*, combining a color anchor with other content; and *abstract idiosyncratic*, using metaphorical or affective expressions. They are coarse parser-based categories, not exhaustive semantic analyses. Examples and counts appear in Appendix A.

3.2 Interactive states and evaluation

In the original protocol, the guesser initially observes only x_i and outputs HEX. The teacher sees the target and current guess, emits feedback, and the guesser receives the description, previous guess, and latest message. The reference game permits up to three revisions, with early stopping at $\Delta E_{76} < 5$. Controlled studies below either supply the start or replace description prediction with an isolated adjustment task.

For direct-LAB outputs, retain both native coordinates z_t and the displayed state $\hat{c}_t = P(z_t)$, obtained by conversion to clipped, rounded uint8 sRGB and back to LAB. Feedback and subsequent prompts use the displayed state in the matched controls. Native and displayed behavior can differ: rectangular LAB encoding bounds are not the sRGB gamut.

3.3 Metrics

The primary error is $e_t = \|c - \hat{c}_t\|_2 = \Delta E_{76}$. We report initial and endpoint error, gain $e_0 - e_T$, and mean per-example relative improvement $(e_0 -$

$e_T)/(e_0 + \epsilon)$; the latter is not the percentage reduction of the cohort mean. Alignment compares the displayed update $u_t = \hat{c}_{t+1} - \hat{c}_t$ with ideal correction $v_t = c - \hat{c}_t$:

$$A_t = \frac{v_t^\top u_t}{\|v_t\|_2 \|u_t\|_2}. \quad (1)$$

Zero-length vectors are excluded from alignment, not from target error. Constraint satisfaction checks signed movement along supplied oracle directions, including HSV saturation where applicable.

For an isolated instruction with unit requested direction d , we measure signed step $s = d^\top(z' - z)$ and off-axis drift $\|(z' - z) - sd\|_2$, also reporting displayed counterparts. Positive steps follow the requested direction; zero and wrong-sign steps remain separate. Ordinal comparisons test the ladder *a little* < *somewhat* < *much*; bare wording is unranked. Projection cost is $\|z' - P(z')\|_2$. Direction compliance alone is insufficient: a nonzero update reduces Euclidean error only when $\|u\|_2 < 2\|v\|_2 \cos \theta$.

4 Experimental Setup

4.1 Oracle feedback and fresh grounding evaluation

Oracle policies compute LAB lightness, red–green, yellow–blue and HSV saturation discrepancies, threshold candidates, and rank them by normalized absolute error. They emit up to one, two, or three selected constraints. The minimal and axis- $c1$ policies share a one-constraint setting; template wording expresses the same selected directions through deterministic phrases. These phrases do not introduce the graded modifiers studied separately below. Examples appear in Appendix B. An additional primitive-direction template variant was originally named *template-vocab*; the implementation does not establish a ceiling for all broader teacher vocabulary.

The fresh Qwen3-14B grounding evaluation uses 1,000 descriptions, 250 per regime, sampled with seed 113 after excluding all 400 original debug-subset IDs. Every game uses HEX output and three fixed revisions with canonical axis feedback supplying at most three constraints. The initial prediction also provides its paired one-shot baseline. All 4,000 responses parse. The sample is equally weighted across regimes, not weighted by corpus frequency. Plans, selected IDs, source/subset hashes and prompts are frozen; the model revision

is pinned to 40c06982. This protocol is a fresh evaluation, rather than reconstruction of the earlier unrecoverable runs.

4.2 Matched output-interface control

On 64 descriptions, 16 per regime, compare HEX, plain LAB, and LAB with an explicit axis legend. Each receives the same displayed start from the parent HEX predictions and the same first oracle correction, with at most three constraints. Three fixed revisions yield 192 games and 576 generations; later adaptive messages can differ. The first revision is the primary matched endpoint, and the third is secondary. Fresh contexts show only the latest feedback. Supplied starts cost no generation within this control.

4.3 Graded wording and held-out control

An isolated probe uses 12 fixed colors, six directions, four wordings (bare, a little, somewhat, much), and plain versus axis-legend LAB prompts. Exact 5-/10-unit and no-change controls separate coordinate execution from qualitative instructions: 888 responses in total, all in fresh contexts.

In the exploratory pilot, 12 separately drawn sRGB calibration colors produce 288 direction/wording responses. Fit direction-specific medians of displayed signed movement, including non-positive parsed movements; only positive graded medians are selection candidates. For a hidden target, choose the phrase whose median is closest to the current requested-axis residual. The bare arm supplies direction only; a numeric control supplies the exact coordinate increment and explicit holds on the other coordinates. These arms share one requested axis, not equal semantic information.

The four-arm confirmation fits medians on 32 new colors (768 responses) and evaluates 128 disjoint starting colors. An unfitted arm selects *a little*, *somewhat*, or *much* using fixed residual cutpoints 9 and 18, without receiver medians. The primary comparison is calibrated minus unfitted displayed target error. Input-only feasibility checks retain 2,108 targets and exclude 196; previous study colors are excluded before inference. Whole starting colors are resampled for intervals. Gemma 4 12B repeats the identical frozen input plan, task order and prompts, fitting its own medians on the same 32 calibration colors. No Qwen responses fit the Gemma policy, and no held-out responses fit either policy.

The pilot’s one-step evaluation uses 12 disjoint

starts and 196 feasible targets. Its sequential transfer freezes the pilot policy and uses 12 further fresh starts, 200 targets, target-known stopping at $\Delta E_{76} \leq 5$, and a five-revision cap. Targets are constructed at native distances 6, 12, or 24 along one of six LAB directions. Input-only feasibility checks exclude 20/216 and 16/216 targets, respectively; generated outputs are never filtered by projection cost. The original axis remains fixed during transfer, but the requested sign can reverse after overshoot. Appendix C details selection, projection, and cost accounting.

4.4 Teacher restatement and feedback reception

From the shared-start cohort, four input-selected demonstrations (one per regime) leave 60 evaluation examples. Compare the archived natural-paraphrase prompt with a restricted exact-direction prompt, each zero-shot and four-shot, at maximum bandwidths one and three: 480 teacher messages. All conditions see identical target/guess states and oracle directions; the clause budget always covers the supplied count. Demonstration answers are canonical oracle sentences, not teacher outputs selected by performance. This tests oracle-assisted restatement, not unaided geometric generation.

A separate receiver study passes every original teacher message unchanged, plus a canonical oracle arm, to the same HEX guesser from identical starts: 120 cases and 600 single revisions. No audit labels select or repair messages; target coordinates and direction JSON are not appended to receiver prompts. We keep frozen lexical audits, supplementary scope-aware interpretation, and receiving target error separate.

4.5 Models, uncertainty, and failures

Grounding and diagnostic controls use Qwen/Qwen3-14B (Yang et al., 2025); the four-arm magnitude study additionally uses google/gemma-4-12B-it (Gemma Team, 2026), pinned to revision 707f0a3b. Both run through Hugging Face Transformers (Wolf et al., 2020), bfloat16 on an A100 40 GB, temperature zero, and thinking disabled. Gemma uses its multimodal processor/model loader with text-only inputs. Receiver budgets are 64 output tokens; teacher budgets are 80. Plans, prompts, model-revision metadata, and atomic response checkpoints are preserved. Strict parse failures remain failures, without repaired colors;

Regime	Initial ΔE	Final ΔE	Gain
Explicit	36.41	22.02	14.39
Prototype	43.39	22.41	20.98
Compound	37.89	20.64	17.24
Abstract	51.64	23.51	28.12
Overall	42.33	22.15	20.19

Table 1: Fresh Qwen3-14B grounding evaluation: 250 descriptions per regime, canonical c3 feedback and three fixed revisions. Initial predictions are the paired one-shot baseline. Gain is initial minus final error.

backend exceptions remain pending. Grounding, shared-start, and teacher-reception outputs parse fully. The four-arm confirmation has 10 held-out parse failures and one calibration failure; pilot calibration parses 287/288 responses. Gemma calibration parses 768/768; its 27 held-out failures all occur in the numeric arm. All three Gemma language arms parse fully. Failures remain visible in completion counts and matched-cohort denominators.

Paired percentile bootstrap intervals resample whole descriptions within regimes for grounding, whole examples within regimes for interface/teacher reception, and whole starting colors for magnitude control, retaining all associated cases and pooled weights. They condition on observed cohorts, not generation randomness or unrestricted population generalization. Subgroup analyses are exploratory and not multiplicity-adjusted.

5 Results

5.1 Interaction localizes recorded targets

Across all 1,000 fresh descriptions, mean initial error is 42.33 [40.70, 44.01] and final error is 22.15 [21.28, 22.99]. The paired initial-minus-final gain is 20.19 [18.61, 21.79] ΔE_{76} , reducing mean error by approximately 47.7%. This reduction of means is not mean per-example relative improvement. Initial and final convergence counts are 5 and 38 of 1,000 at the $\Delta E_{76} \leq 5$ threshold.

All regimes improve descriptively (Table 1). Abstract descriptions have both greater initial disagreement and larger absolute gain; these means do not establish a separate between-regime effect or semantic implausibility of alternative colors. This evaluation measures localization after additional corrections; it does not isolate sequential adaptation from feedback information at a matched constraint budget.

Interface	First ΔE	Third ΔE	C-sat
HEX	32.10	22.78	0.888
Plain LAB	38.29	35.28	0.823
LAB + legend	32.58	23.16	0.906

Table 2: Shared starts and first corrections, $N = 64$ per interface; mean starting error 41.96. Displayed-state error is primary. Three revisions are forced rather than stopped at convergence.

5.2 Output encoding affects correction from shared starts

All 576 shared-start revisions parse; every interface is evaluated on all 64 examples (Table 2). Plain LAB exceeds HEX first-revision error by 6.19 [0.85, 11.89] ΔE_{76} . Adding an axis legend reduces that error by 5.71, but its first-revision interval spans zero. After three revisions, legend minus plain LAB is -12.11 $[-18.04, -5.49]$. Thus, the encoding/instruction intervention affects feedback following beyond differences in initial prediction. Legend and HEX endpoints are close, but their difference 0.38 $[-4.85, 6.29]$ does not establish equivalence. Later feedback is adaptive, and the result is conditional on HEX-derived starts.

5.3 Graded instructions specify unequal step sizes

All 888 isolated-probe responses parse. Explicit axis instructions increase positive-direction qualitative updates from 228/288 to 288/288. Strictly increasing ordinal comparisons rise from 172/216 to 210/216; nondecreasing comparisons, which include ties, rise from 208/216 to 216/216. These overlapping within-color comparisons concern 12 fixed anchors, not 216 independent colors. Numeric and no-change controls are exact except one legend-condition 10-unit instruction executed as a 20-unit update.

The exploratory pilot calibration split gives displayed median steps for *a little* ranging from 1.14 to 4.92 across directions, *somewhat* from 5.07 to 10.09, and *much* from 28.72 to 75.84. These are model/prompt-specific operational measurements, not universal perceptual units. Large updates often leave the displayable gamut: 54/72 calibration responses for *much* have projection error above one. Native step variation and display projection must be distinguished; neither the model’s internal interpretation nor human perceptual nonuniformity is identified by these outputs alone.

5.4 Calibrated magnitude improves held-out localization

Each model completes all 8,432 held-out responses (Figure 2). For Qwen, there are 2,098 fully parsed quartets spanning all 128 starting colors; parse failures number 3, 5, 0, and 2 for bare, unfitted, calibrated, and numeric feedback. Calibrated wording reduces mean target error from 12.36 to 6.86 relative to the unfitted rule: paired difference -5.50 $[-5.79, -5.20]$. The available-pair contrast is similar (-5.53). Unfitted versus bare error differs by 0.45 $[-0.69, 1.52]$, so graded wording alone does not establish improvement in this comparison. Both graded arms follow the requested direction on approximately 99% of parsed responses. Direction compliance therefore does not ensure accurate magnitude control. This tests the fitted policy against one fixed rule; it does not establish that every unfitted rule is inferior.

Exploratory saved-output breakdowns localize the aggregate benefit. At requested distances 6, 12, and 24, calibrated-minus-unfitted error is -0.71 , 0.00, and -18.35 , respectively. The largest reductions occur for 24-unit red, yellow, and blue corrections; the 24-unit calibrated mean error remains 15.31. Harmful updates fall from 318 to 58 on the common cohort. However, 1,419 cases tie within 0.01, versus 556 wins and 123 losses, and the median paired difference is zero. Calibration chiefly avoids excessive large corrections in this task rather than improving every adjustment. Subgroup intervals are exploratory and the prompt-identity audit shows that 1,385 common-cohort cases receive identical prompts and produce identical colors. Policies differ in the remaining 713 cases: calibration selects *somewhat* instead of *much* for 24-unit red, yellow, and blue targets, and instead of *a little* for 6-unit lighter, red, and green targets. All 12-unit and darker comparisons reuse identical prompts. These observations localize the policy differences; they are not an independent controller experiment.

Gemma replicates the primary contrast: on 2,081 common parsed quartets, calibrated minus unfitted error is -5.16 $[-5.48, -4.84]$. The language-only available-pair contrast includes all 2,108 cases and is -5.10 $[-5.42, -4.77]$, so the benefit persists without excluding numeric parse failures. Bare wording is also better than the unfitted rule by 2.69 units; a quantifier ladder alone is not sufficient. The two models’ main means use different parsed co-

Study	Arm	Final ΔE	≤ 5	Calls
Qwen	Bare	11.91	–	2,108
	Unfitted	12.36	–	2,108
	Calibrated	6.86	–	2,108
	Numeric	0.35	–	2,108
Gemma	Bare	8.36	924	2,108
	Unfitted	11.06	1,400	2,108
	Calibrated	5.89	1,398	2,108
	Numeric	0.08	2,081	2,108
Sequential	Bare	6.46	148	516
	Calibrated	2.98	192	338
	Numeric	0.10	200	205

Table 3: One-step means on model-specific common quartets: 2,098 for Qwen and 2,081 for Gemma, each spanning the same 128 starts. Calls include failed outputs and exclude 768 calibration calls per model. Sequential Qwen pilot: 200 targets from 12 further starts, stopping at error ≤ 5 with a five-revision cap, excluding 288 calibration calls. Numeric feedback supplies additional precision.

horts and are not a direct ranking of model ability.

Gemma’s exploratory distance effects are -0.06 , 0.00 , and -18.18 at 6, 12, and 24 units. Of its 2,081 common cases, 1,524 share identical prompts and colors. The other 557 contain 443 wins, four ties and 110 losses. Calibration replaces *much* with *somewhat* for 24-unit darker, red, yellow and blue corrections, and usually replaces *a little* with *somewhat* for 6-unit green corrections. Harmful updates fall from 333 to 75, but convergence counts do not increase (1,400 versus 1,398). Every observed starting-color mean favors calibration, and leave-one-start-out effects range from -5.19 to -5.12 . These descriptive audits support lower average error and fewer harmful updates, rather than universal improvement or greater threshold success.

The Qwen exploratory pilot’s transfer to 12 further starts completes all 600 games with 1,059 parsed revisions. Calibrated minus bare endpoint error is -3.48 [-4.20 , -2.74]; convergence rises from 148/200 to 192/200. Evaluation calls fall from 516 to 338 (34.5%), excluding calibration. All twelve per-start mean contrasts favor the calibrated policy, but individual targets do not uniformly improve. This sequential pilot has no unfitted graded-selection arm; the larger four-arm confirmation tests that contrast for one revision only.

Eight calibrated trajectories exhaust the cap: all concern 24-unit lighter or greener targets and end with off-axis disagreement exceeding the threshold. The fixed-axis policy does not explicitly restore display-induced drift. Numeric feedback has the

Teacher arm	Matches	ΔE	C-sat
Natural, 0-shot	81	36.89	0.650
Natural, 4-shot	120	34.40	0.863
Restricted, 0-shot	104	31.94	0.867
Restricted, 4-shot	120	33.68	0.868
Oracle	120	33.78	0.868

Table 4: Oracle-assisted direction restatement and one-revision reception, 120 cases on 60 examples. Matches are supplementary resolved set matches; natural zero-shot has 22 unresolved cases. All receiver endpoints remain in the comparison, with common starting error 39.83. C-sat uses supplied oracle directions, not inferred teacher intent.

lowest endpoint error, but retains execution tails: four one-step responses have native error above one (maximum 39), and five sequential responses exceed one (maximum 44). The latter reach convergence after subsequent oracle corrections, not autonomous self-repair. Appendix D preserves these limitations.

5.5 Restatement fidelity and usefulness are distinct

All 480 teacher messages and 600 receiver revisions complete. A supplementary scope-aware audit resolves all supplied-direction sets in both four-shot conditions as preserved; restricted zero-shot preserves 104/120. Natural zero-shot has 81 resolved matches, 17 mismatches, and 22 unresolved cases (Table 4). These are finite-grammar diagnostics, not independent human judgments. Literal keywords can invert the intended correction: *your guess is darker than the target* implies a lighter update. Harmless *Feedback*: prefixes can also lower strict format certification.

Restricted zero-shot reduces receiver error relative to natural zero-shot by 4.95 [1.32, 8.39]. The natural four-shot mean improvement is 2.49, but its interval includes zero. Faithful restatement therefore does not guarantee the best receiving accuracy. Restricted four-shot is close to the oracle (difference -0.10 [-0.40 , 0.09]), without establishing equivalence.

Restricted zero-shot’s advantage over the oracle is concentrated: 11 wins, 104 ties, 5 losses, and median difference zero. Its 98 literally identical messages tie; six changed same-set messages and sixteen messages with additions contribute 57.1% and 42.9% of its net advantage. This is an arithmetic decomposition, not a causal effect of surface wording or extra information. Four of the six same-

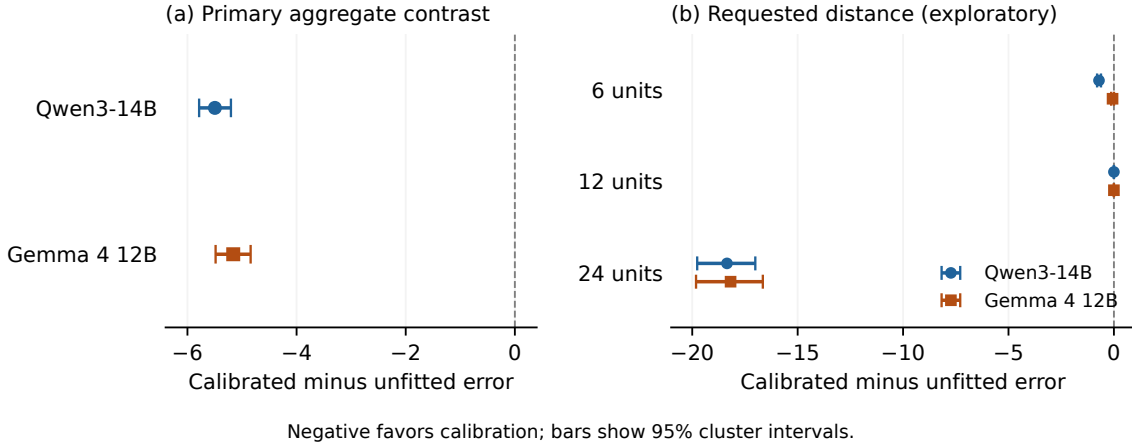


Figure 2: Receiver-specific calibration reduces displayed target error relative to the fixed unfitted quantifier rule for both models (left). Exploratory distance breakdowns concentrate the benefit at 24-unit targets (right). Bars show 95% starting-color cluster percentile intervals; each model covers the same 128 starting colors, with 2,098 Qwen and 2,081 Gemma common parsed quartets. These are within-model contrasts, not a paired between-model test. The panels use different horizontal scales. Values are reproduced from reported rounded estimates.

set messages reorder directions; two change punctuation or conjunctions. One reordered message contributes 92.01 of their net 126.04 summed error advantage, showing concentration in a single case rather than a consistent wording benefit. The sixteen additions also prevent an information-matched claim of better restatement. These controls establish prompt sensitivity in oracle-assisted feedback, not a general inability to generate geometrically correct guidance unaided.

6 Discussion

Language as a graded control interface. Direction, magnitude, and stopping are separate requirements. Given $\Delta E_{76} \leq 5$, further valid-axis corrections can still overshoot. A saved-prefix stopping replay retains nine otherwise lost convergence outcomes and saves 23/576 revisions in the shared-start study (Appendix D.3); this is target-known stopping, not an available model-only policy. The two-model magnitude studies show that calibration lowers average error under the tested fixed rule, chiefly for large requested corrections. This need not increase threshold success: Gemma’s common-cohort convergence counts remain nearly unchanged despite fewer harmful updates. Display projection and gaps in a three-phrase ladder still limit accuracy.

Grounding versus coordinate execution. Shared-start results isolate correction from initial name prediction; isolated numeric probes further

separate execution from linguistic magnitude. LAB with a legend is an instructed coordinate interface, not proof of unaided perceptual understanding. The observed color control is relevant to creative tools, but its utility in image editing, where descriptions and transformations are richer, requires separate evaluation.

Teacher claims require matched evidence. Comparative framing, restricted instructions, and demonstrations affect restatement. A word-level audit is not a semantic judgment, and preserving directions is not equivalent to maximizing target-error reduction. Our conclusions concern tested prompt pipelines and receiving behavior, not a universal following-versus-generating capability gap.

7 Conclusion and Future Work

COLORREF evaluates feedback-driven color prediction beyond a one-shot endpoint. Controlled feedback improves target localization; matched controls expose encoding sensitivity, direction-dependent graded steps, and the distinction between teacher restatement and usefulness. Independently calibrated wording policies lower held-out one-step error for two model families relative to the same fixed quantifier rule, primarily by avoiding excessive large corrections. A smaller Qwen sequential pilot also reduces endpoint error and evaluation calls; this cost benefit has not been replicated with Gemma. Projection drift, coarse phrase resolution and numeric output failures remain. Fu-

ture work should test alternative target distributions, human and multilingual interaction, cross-model policy transfer, multi-axis control, and downstream creative editing.

8 Limitations

Our recorded crowd targets need not be uniquely appropriate for their descriptions. Both the retained corpus and raw English source record one color per description, without repeated targets for estimating agreement or acceptable variation. Validity filtering is not target-agreement measurement; alternative-color judgments or target distributions remain needed, especially for abstract names. CIELAB is approximately perceptually uniform, but Euclidean ΔE_{76} is not a human magnitude calibration or a strict visibility threshold. Equal coordinate steps have equal norm in this metric; differences between directions can also reflect generated native steps and the restricted sRGB gamut, rather than perceptual nonuniformity alone.

The magnitude confirmation covers two model families with the same prompt and 128 held-out starting-color clusters. Other diagnostic controls use Qwen alone and smaller fixed samples, including twelve-color pilot clusters. The primary comparator is one fixed quantifier rule with cutpoints 9 and 18; alternative unfitted rules and prompt paraphrases are not tested. The studies use English descriptions and supplied numeric or HEX-derived starting states. Bootstrap intervals do not establish broad population generalization, generation-randomness uncertainty, or multiplicity-adjusted significance. Failed calibration and held-out outputs remain in counts; matched parsed cohorts can introduce selection bias. No outputs are repaired or filtered by projection cost.

Teacher studies test oracle-assisted restatement, not unaided geometry; supplementary finite-grammar and assistant review are not independent human labels. Restriction changes several prompt instructions simultaneously. Numeric feedback adds exact magnitude and coordinate holds, and calibrated wording adds a selected magnitude category; neither is information-matched to bare directions. Target-known stopping and benchmark oracles are not deployable model-only policies. Evaluation-call savings omit calibration cost, and fixed-axis transfer does not intentionally correct off-axis drift. Creative applications, multilingual behavior, HSV magnitude calibration, and alterna-

tive perceptual metrics require further studies.

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A Corpus Construction and Examples

The ColorNames-derived source pairs user-provided color names with sRGB HEX codes. Pre-processing retains English strings with valid HEX codes and a description-validity score ≥ 0.75 . The original validity scoring uses GPT-4o to judge whether a string plausibly functions as a color description, accepting literal, figurative, and subjective expressions while rejecting spam, URLs, non-sensical strings, and personal messages. This score is not a measure of inter-speaker agreement about the paired color. Canonicalization assigns stable identifiers and normalizes fields; color conversion produces RGB, HSV, and CIELAB coordinates. The final corpus has 518,830 rows.

A read-only inventory confirms that 43,187 retained rows have score exactly 0.75; a strict > 0.75 filter would retain 475,643 rows instead. Each of the 705,869 raw English entries and 518,830 retained entries has a distinct description, including after lowercasing and whitespace normalization. Neither inventory contains repeated description–color pairs or multiple recorded colors for a description. The 725,203 evaluation records likewise have distinct exact descriptions; their fields concern language, reasoning, and quality score, without explicit votes or alternative color targets. These inventories establish recorded multiplicity, not human agreement: evaluation therefore measures distance from the assigned recorded target, rather than from a uniquely correct interpretation.

Regime	Count	Percent
Abstract idiosyncratic	238,416	46.0
Prototype mediated	137,420	26.5
Compound associative	94,246	18.2
Explicit grounded	48,748	9.4

Table 5: Original corpus regime counts; rounded percentages need not sum to exactly 100. Labels are coarse analysis categories.

Regime	Description	Recorded target
Explicit	faded old rose	#D2917A
Prototype	montmorency cherry	#E13D2B
Compound	cryogenic electric blue	#1926FF
Abstract	woman of the dunes	#66615E

Table 6: Illustrative descriptions and their recorded targets, retained from the original manuscript. A target is not asserted to be the only semantically plausible color.

Structural parser. The deterministic parser lowercases strings, maps punctuation to spaces, removes nonalphabetic characters, and collapses whitespace. It scans tokens right-to-left for a basic-color head; preceding tokens are partitioned into known modifiers and unknown prefix content. Confidence decreases with unexplained prefixes. Parser v2 additionally normalizes selected *-ish* forms and plurals, allows prototype heads when no basic head is found, and records prototype hits throughout the name. Hand-curated YAML lexicons and explicitness, prototype, rarity, and abstraction scores determine the regime.

B Oracle Wording and Receiving Prompts

The subscript $c1$, $c2$, or $c3$ specifies a maximum constraint count, not a magnitude. For target #A0B0C0 and guess #805000, the implementation selects muted, lighter, and blue in that order. These examples are generated by the available code with example ID 0, seed 13, and feedback turn 0; they are not recovered historical experiment messages.

Condition	Example feedback
Minimal $c1$	Make it more muted.
Axis $c1$	Make it more muted.
Axis $c2$	Make it more muted and lighter.
Axis $c3$	Make it more muted, lighter, and more blue.
Template $c1$	Make it less saturated.
Template $c2$	Make it less saturated. Move toward a brighter shade.
Template $c3$	Make it less saturated. Move toward a brighter shade. Move it toward blue.

Template conditions use deterministic alternative phrases for the same selected directions; the phrase bank does not add *a little* or *much*. These modifiers belong to the separate magnitude studies. Minimal and axis- $c1$ are equivalent in this controlled setting. The template-vocab implementation still selects primitive LAB/HSV directions; broader warmer/cooler or gray dictionary entries do

not establish a matched broad-vocabulary ceiling.

No-candidate fallback messages can differ.

The common HEX revision template is:

You are playing a color guessing game.

Color description:

"{raw_name}"

Your previous guess was:

{previous_guess_hex}

Feedback: {feedback}

Revise your guess.

Answer only with a valid hex

code in the form ##RRGGBB.

The initial HEX template omits the previous guess and feedback and asks for the intended color. New receiving studies use unchanged teacher text in the feedback field. The guesser does not receive the target or oracle constraint JSON as additional prompt fields. All contexts are fresh.

Plain LAB revisions instead request LAB (L , a , b) within $L \in [0, 100]$ and $a, b \in [-128, 127]$. The legend explicitly states that increasing/decreasing L makes the color lighter/darker, increasing/decreasing a shifts toward red/green, and increasing/decreasing b shifts toward yellow/blue. Previous displayed LAB coordinates have six decimal places. These rectangular bounds are encoding bounds, not the sRGB gamut boundary. The model must supply three numeric values; unevaluated expressions and incomplete triples are not repaired.

C Matched Controls and Reproducibility

C.1 Shared-start interface control

The 64 descriptions are balanced across regimes. Starts come from the completed parent HEX predictions, not new predictions by each receiving interface. Target LAB is recomputed from recorded target HEX. The same oracle thresholds and ranking operate on displayed uint8 sRGB states in every arm. First corrections are identical; later corrections respond to the arm's own displayed state. Three revisions are forced even after a close guess, which differs from the original early-stopping protocol. All 576 generated revisions parse, so first/final matched cohorts contain all 64 examples. First alignment denominators are 64/61/63 for HEX/plain LAB/legend LAB because some updates have zero length.

Final native errors are 22.781, 35.589, and 23.325; displayed errors are 22.781, 35.275, and 23.161. Mean projection errors over generated states are 0, 0.783, and 0.937. Thus, an axis legend does not uniformly reduce projection cost. These results must not be mixed with the earlier own-initial-prediction interface study, whose common cohort is 61/64 after three legend-condition parse failures.

C.2 Isolated wording probes

For each of 12 fixed colors and six requested directions, compare bare, *a little*, *somewhat*, and *much* under plain and legend prompts. Each response is independent; the ordinal comparison does not chain the resulting states. Exact 5-/10-unit controls explicitly name a coordinate and hold the other two fixed. Twelve no-change cases per prompt complete the 888-response plan. All responses parse. The qualitative ladder has three pairwise comparisons per starting-color/direction combination, 216 comparisons per prompt. Bare wording has no ordinal rank.

Measure	Plain	Legend
Positive qualitative updates / 288	228	288
Nondecreasing pairs / 216	208	216
Strictly increasing pairs / 216	172	210
Tied pairs / 216	36	6

Table 7: Direction and ordinal magnitude in isolated probes. Nondecreasing includes ties and does not imply correct-sign movement.

There is no hidden target for qualitative probes. A numeric headroom fraction uses the prompt’s rectangular bounds and cannot measure a fraction of remaining target distance or gamut capacity. Projection diagnostics are not response filters. The one nonexact ordinal-study numeric control returns a 20-unit yellow-axis update for an instruction specifying ten.

C.3 Calibration policy and target construction

Calibration and one-step evaluation use 12 colors each drawn from an inclusive sRGB channel range [48,208], with seed 29. A subsequent seed-47 transfer split contains twelve fresh starts, excluding the earlier calibration/evaluation starts, isolated anchors, and the inspected close-case start by exact HEX. Inputs and feasibility decisions are frozen before inference.

Calibration generates 12 colors $\times$ six directions $\times$ four wordings, or 288 responses (287 parsed). For each direction and phrase, take the median displayed signed step among parsed responses, retaining zero/wrong-sign updates. Bare wording is not a selection candidate; graded phrases with positive medians are. Select the median closest to the current axis residual, breaking ties by smaller median and then ladder order. No held-out response refits this mapping.

Direction	A little	Somewhat	Much
Lighter	2.949	7.961	28.722
Darker	4.001	10.013	30.118
More red	1.139	5.073	75.844
More green	1.277	6.620	30.038
More yellow	4.529	7.381	54.333
More blue	4.917	10.094	48.137

Table 8: Frozen calibration medians in displayed requested-axis units. These are operational, direction-specific measurements for this model/prompt, not human or universal perceptual magnitudes.

Native target candidates move one coordinate by 6, 12, or 24 units. Retain candidates within the prompt bounds whose projection error is at most one and whose displayed residual follows the intended direction. This produces 196/216 one-step targets and 200/216 transfer targets; excluded inputs are not replaced. Generated outputs remain unfiltered. Displayed target construction and six-decimal prompted coordinates are stored separately from exact instructed targets.

All arms use the same supplied start and target. Bare feedback specifies direction. Calibrated feedback supplies a selected magnitude phrase. Numeric feedback specifies an exact increment and holds the other coordinates; it is a precision/execution control, not information-matched wording. Sequential transfer retains the original axis throughout, stops at the first displayed error at most five, and caps generation at five revisions. If the axis residual is at most 0.01 while total error exceeds five, the protocol permits an explicit off-axis-residual ending instead of an empty correction. No such ending occurs in the completed transfer study. Reaching the cap is a valid observed endpoint, not an imputed success.

C.4 Four-arm magnitude confirmation

The confirmation uses seed 211, 32 new calibration colors and 128 held-out starting colors, excluding

49 previously used colors before inference. Calibration covers six directions and four wordings (768 responses, 767 parsed). Frozen direction-specific medians use parsed responses including nonpositive updates; held-out responses never refit the mapping. The fixed unfitted rule selects *a little* below residual 9, *somewhat* from 9 to below 18, and *much* otherwise. Calibrated wording instead selects the positive graded median nearest the requested-axis residual. Bare directions and exact numeric instructions complete the four arms. All use fresh contexts and one revision.

The existing native-distance targets at 6, 12 and 24 units yield 2,108 feasible cases; 196 candidates fail frozen input-only feasibility checks. All 8,432 held-out responses complete. Parse failures are 3/5/0/2 for bare/unfitted/calibrated/numeric. Primary means use 2,098 common parsed quartets covering all 128 starts. The primary calibrated-minus-unfitted contrast is -5.497 [$-5.786, -5.203$]; available paired responses give -5.528 [$-5.814, -5.245$] on 2,103 cases. All 12-unit and darker comparisons share identical prompts. On the common cohort, 1,385 prompt-identical cases produce identical colors; the other 713 cases contain 556 wins, 34 ties and 123 losses at tolerance 0.01. This identity partition is descriptive, not a replacement primary cohort. Numeric execution has 39 parsed native errors above one, maximum 51.004; misses remain in the reported performance.

C.5 Gemma replication and saved-output diagnostics

Gemma 4 12B uses revision 707f0a3b8a3c7ad586ed01e27eafbad8a27dd0f7, the identical frozen plan and text prompts, and independently fitted medians. All 768 calibration outputs parse. Of 8,432 evaluation responses, only 27 numeric outputs fail parsing; bare, unfitted and calibrated each parse 2,108/2,108. Common quartets contain 2,081 cases from all 128 starts. The primary contrast is -5.161 [$-5.484, -4.842$]; available language pairs give -5.095 [$-5.417, -4.772$] over all 2,108 cases.

Direction-specific medians differ between receivers under this protocol: for *a little more red*, displayed medians are 1.054 for Qwen and 4.953 for Gemma. The experiment does not test a controller fitted on one model and deployed on the other, so it establishes within-model calibration

benefits, not cross-model policy transfer.

The Gemma identity audit finds 1,547 identical prompts among planned unfitted/calibrated pairs, with no parsed-LAB mismatch on identical prompts. On common quartets, 1,524 identical prompts give zero error difference; 557 changed prompts give a mean difference of -19.282 . The overall median paired difference remains zero. Distance-specific effects at 6/12/24 units are $-0.064, 0.000$, and -18.182 ; the largest gains occur for 24-unit red, yellow and blue targets. Harmful updates fall from 333 to 75, while threshold success remains nearly unchanged (1,400 to 1,398). All 128 per-start mean differences favor calibration; omitting one start yields differences between -5.194 and -5.121 . These audits are exploratory and do not replace the primary cohort.

Among parsed numeric outputs, mean native execution error is 0.006; 2,058/2,081 are within 0.01 of instructed coordinates. The 27 invalid outputs remain failures, without imputed colors or repaired responses. Saved-output inspection finds all 27 failures in numeric *more green* instructions (decreasing *a*), across 17 starts: four at distance 6, seven at 12 and sixteen at 24. Seventeen outputs contain only two coordinates; ten contain a triplet rejected for negative *L*. Twenty-five terminate at EOS below the budget, and two reach the 64-token cap after first emitting an invalid triplet and starting a correction. Thus, truncation does not explain most observed failures.

For example, starting from (49.742386, 10.289397, 5.133036) and decreasing *a* by 24.238567 requires (49.742386, -13.949170 , 5.133036); the model instead returns LAB(-13.94917 , 5.133036). This example computes the requested *a* value but omits *L*. Other outputs place a negative adjusted value in the *L* position or duplicate coordinates. These are observed coordinate-output failures, not evidence of a particular internal reasoning mechanism. No missing coordinate is supplied retrospectively. Numeric parsing succeeds on 2,081/2,108 planned responses (98.7%); its low parsed-response error does not establish reliable execution across all planned cases.

C.6 Uncertainty and saved inputs

Paired bootstrap draws resample whole starting colors for magnitude studies and whole examples within regimes for interface/teacher studies,

keeping associated direction/distance or bandwidth cases together. Analysis settings are retained with the saved run metrics. Pooled case weights remain unchanged. The pilot uses twelve starting-color clusters; the confirmation uses 128. Multiple targets per start are not independent colors. Available-pair sensitivity equals common-triplet estimates in the fully parsed pilot; confirmation quartets and available pairs have different, explicitly reported denominators.

Frozen configuration, plans, calibration/controller snapshots, rendered prompts, responses, parsing results, native/displayed states and scores are retained. Resume validates hashes, scores and observed model revision; backend failures stay pending, while invalid generated outputs remain failures. The observed model snapshot is recorded in run metadata. CPU inspectors write supplementary reports without changing checkpoints or fitting policies. Software tests use simulated outputs and do not supply model-performance evidence.

D Magnitude and Stopping Diagnostics

D.1 One-step limitations

Calibrated versus bare has 93 wins, 57 ties and 46 losses on 196 targets at an exploratory 0.01 ΔE tie tolerance. Mean differences for 6-/12-/24-unit targets are -9.84 , -5.77 , and $+0.85$, respectively; the 24-unit interval spans zero. The red calibration ladder has a large gap between *somewhat* (5.073) and *much* (75.844) displayed units. This diagnoses coarse resolution, not an identified internal linguistic mechanism. Per-start leave-one-out mean effects remain favorable, but are descriptive leverage checks rather than new intervals or grounds for exclusion.

The exact numeric control has 188/196 native responses within 0.01 of the instructed coordinates and 192 within one. Four larger misses are 39, 32, 12, and 10 units. They terminate at EOS below the output budget and preserve unrequested coordinates, with projection displacement below 0.5. They remain in reported performance. For example, an instructed red-axis change from -43.698223 to -37.939764 instead returns 1.060236, a 39-unit miss. Observed execution/transcription failure does not prove a particular internal digit or sign-loss mechanism.

D.2 Sequential prefixes and failures

All rows below concern the same 200 transfer targets. Earlier stopped states are retained with no additional calls. Prefixes are saved observations, not independently rerun or optimized stopping policies.

Budget	Bare	Calibrated	Numeric
1	10.147	6.434	0.534
3	7.006	3.339	0.095
5	6.461	2.979	0.095

Table 9: Mean displayed target error at observed transfer prefixes under the frozen five-revision protocol.

Calibrated convergence counts are 119, 187, and 192 at these budgets. The fourth/fifth revisions add five successes at 22 additional calls; the primary endpoint remains the declared five-revision cap. Bare/calibrated trajectories contain 152/11 harmful revisions and 174/48 requested-sign reversals. Their unequal adaptive histories make these descriptive counts, not independent per-step significance samples.

All eight calibrated cap endings concern 24-unit lighter/greener corrections on six starting colors. Their final off-axis disagreement exceeds five; seven also have requested-axis residual below five. Large native proposals project into shifted unrequested coordinates. The fixed-axis controller does not explicitly repair that drift; it is not universally impossible for later projection to change unrequested coordinates incidentally. Bare converges on six of these eight targets and numeric on all eight.

Numeric transfer has 199/205 native revisions within 0.01 and 200 within one, with maximum native error 44. Five large first-revision errors reach the threshold after a second target-known correction; one further small error is below the stopping threshold. Endpoint success cannot erase intermediate execution failures or establish autonomous self-correction.

D.3 Target-known stopping replay

Replay the shared-start trajectories, retaining the first displayed state with $\Delta E_{76} \leq 5$, including assigned starts. This performs no new generation and does not select the retrospective best state.

Interface	e_T	e_{stop}	n_T	n_{stop}
HEX	22.781	22.005	192	185
Plain LAB	35.275	34.562	192	186
LAB + legend	23.161	22.822	192	182

Table 10: Stopping replay on the same 64 examples in each interface. Columns e and n give mean target error and generation calls, respectively, for fixed and stopped policies.

Nine otherwise lost convergence outcomes occur on seven distinct examples; three share the initially close *dull toy red* start. Its starting error is 2.380, yet valid-axis first corrections produce gains of -27.646 , -47.017 , and -33.298 in HEX/plain LAB/legend LAB. These are normally terminated outputs with low projection cost: magnitude, not simply geometric sign, can destroy reached success. This single case is not a near-target failure-rate estimate. Stopping is benchmark target-known, not a learned or deployable model-only policy.

E Teacher Prompts, Audits, and Reception

The natural zero-shot condition preserves the archived oracle-assisted prompt: it supplies correct directions as JSON and asks for a natural short comparative paraphrase. Its vocabulary includes LAB/HSV primitives as well as warmer/cooler and gray terms. The restricted prompt instead requests exactly the supplied primitive directions, prohibits additions, omissions, reversals, substitutions and magnitude modifiers, and requires a single *Make it* sentence. This changes multiple instructions and the allowed format simultaneously; it does not isolate one individual wording instruction.

Four-shot prompts use canonical oracle answers for four input-selected demonstrations. IDs are 202953, 207855, 39272, and 352171; evaluation excludes these IDs and normalized description or target/guess-pair overlap. The same IDs and order appear in both forms, with bandwidth-specific answers. The final 60 examples contain 15 per regime; each is evaluated at $c1$ and $c3$. The clause budget equals the maximum supplied direction count and never forces lossy compression. Demonstrations do not cover every direction: darker and more saturated are absent from their logged primitive coverage.

All teacher conditions use Qwen3-14B, temperature zero, thinking disabled, and an 80-token output cap; the HEX receiver uses the same model family

and a 64-token cap. Four-shot prompts necessarily cost more input tokens. The five receiving arms use every message unchanged, with no target-HEX mentions observed and no parse failures in 600 revisions. There is no new teacher generation for the deterministic oracle arm.

E.1 Frozen lexical audit versus scope-aware diagnostics

Primary conservative certification requires an exact detected direction set, without unknown or ambiguous wording or duplicate mentions. It is an operational format/vocabulary criterion, not semantic ground truth. Supplementary finite-grammar auditing resolves explicit guess-versus-target comparisons and a leading *Feedback*: label while retaining unresolved language for review. It does not overwrite primary scores.

Condition	Certified	Match	Review
Natural, 0-shot	7	81	22
Natural, 4-shot	3	120	0
Restricted, 0-shot	104	104	0
Restricted, 4-shot	105	120	0

Table 11: Counts out of 120 teacher cases; Match denotes resolved matches and Review denotes unresolved cases. Certification and resolved-match scores are different measures; failures of certification are not necessarily wrong corrections. Natural zero-shot also has 17 resolved mismatches.

For example, the response *Your guess is darker than the target color* contains *darker* but implies a lighter correction. In contrast, *Feedback: Make it lighter* directly preserves that direction while violating strict canonical formatting. Contextual assistant review of the 22 unresolved cases is retained separately; it is neither independent human annotation nor a replacement for the finite-grammar or primary metrics. No review label selects teacher messages for receiving evaluation.

E.2 Paired receiving effects and saved-message decomposition

All receiver contrasts retain 120 cases on 60 whole-example clusters, with both bandwidths together. Selected teacher-minus-oracle paired effects are given below; intervals are exploratory and not equivalence tests.

Arm	Δ error	95% interval
Natural, 0-shot	3.110	[−1.491, 7.122]
Natural, 4-shot	0.619	[−1.794, 2.686]
Restricted, 0-shot	−1.841	[−3.772, −0.322]
Restricted, 4-shot	−0.104	[−0.397, 0.085]

Table 12: Displayed target-error differences; negative favors the teacher arm.

Restricted zero-shot versus oracle partitions into 98 identical messages (all ties), six different messages preserving the lexical direction set (sum error difference −126.040; 3 wins, 1 tie, 2 losses), and sixteen messages with additions (sum −94.853; 8 wins, 5 ties, 3 losses). Together these yield the observed mean −1.841, but median zero. The 57.1%/42.9% contributions describe net arithmetic effects, not a randomized test of ordering, wording, or added information. The six same-set messages comprise four reorderings (3 wins, 1 loss) and two punctuation or conjunction changes (1 tie, 1 loss); all occur at c3. In example 52032, *Make it darker, more green, and more yellow* produces #003366, whereas the oracle’s *Make it more yellow, more green, and darker* produces #006600. Teacher-minus-oracle error is −92.008, about 73% of the six cases’ net advantage. These saved matched outputs demonstrate sensitivity to equivalent correction forms, but do not identify an internal processing mechanism or establish a generally better direction order.

Eight addition cases use only threshold-qualified extra primitive directions; eight do not. Threshold qualification is neither sufficient for useful magnitude nor necessary for a direction to have a correct sign. Some contradictory primitive messages still improve target error, underscoring the distinction between fidelity and usefulness. Natural-language keyword additions are scope-sensitive and are not semantic bandwidth violations. Across all arms, identical actual prompts never yield different parsed colors in the saved audit; this is observed consistency, not a proof of backend determinism. Restricted four-shot ties the oracle on 118/120 cases.

F AI Assistance Disclosure

We used Claude Code to assist routine experiment management and ablation execution. We also used Codex for code refactoring, test implementation, analysis scripts, and manuscript drafting. Supplementary contextual review of teacher utterances

was AI-assisted and is not independent human annotation; its labels do not replace frozen audit scores or select receiving examples. GPT-4o supplied the original description-validity scores described above. The authors are responsible for checking experimental designs, outputs, analyses, and the final manuscript.